

Norway

SLEIPNER

Volve F

F-4

F-4

Design: F-4

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table (As Installed) @ 54.90m (As Installed)
Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m (As Installed)
Well:	F-4	North Reference:	Grid
Wellbore:	F-4	Survey Calculation Method:	Minimum Curvature
Design:	F-4	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-4			
Well Position	+N/-S	-2.70 m	Northing:	6,478,560.83 m
	+E/-W	-0.19 m	Easting:	435,049.83 m
Position Uncertainty	0.00 m	Wellhead Depth:	91.00 m	Water Depth: 91.00 m

Wellbore	F-4				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	NETICREFERENCE	19.10.2005	-2.55	71.62	50,281

Design	F-4				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-2.70	-0.19	125.41	

Survey Program	Date	30.09.2017			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
184.12	2,696.43	F-4 Drop Gyro 301007 (F-4)	Wellbore Surveyor, stat	Gyro Tool from GD	
2,740.79	2,740.79	F-4 MWD 12 1/4" Hole 1360m to 2740m (F	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
2,760.00	3,510.00	F-4 MWD 8 1/2" Hole (F-4)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-2.70	-0.19	0.00	0.000	0.000	0.000	
184.12	0.10	233.25	184.12	-2.72	-0.22	-0.01	0.078	0.078	0.000	
224.45	0.16	227.58	224.45	-2.78	-0.29	-0.03	0.046	0.045	-4.218	
264.79	0.84	218.68	264.79	-3.05	-0.51	-0.06	0.507	0.506	-6.619	
305.10	1.60	210.39	305.09	-3.76	-0.98	-0.03	0.579	0.566	-6.170	
345.45	2.76	227.92	345.41	-4.90	-1.99	-0.19	0.985	0.862	13.033	
385.80	5.17	242.27	385.66	-6.40	-4.32	-1.22	1.924	1.792	10.669	

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Design:	F-4	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
426.12	7.04	240.40	425.75	-8.46	-8.08	-3.09	1.399	1.391	-1.391
466.40	8.06	224.23	465.68	-11.71	-12.19	-4.56	1.747	0.760	-12.043
506.65	8.85	205.85	505.50	-16.52	-15.51	-4.48	2.088	0.589	-13.699
546.96	8.90	191.34	545.34	-22.36	-17.48	-2.69	1.662	0.037	-10.799
587.28	8.91	181.74	585.17	-28.54	-18.18	0.31	1.104	0.007	-7.143
627.54	9.00	180.61	624.94	-34.81	-18.31	3.83	0.147	0.067	-0.842
667.63	9.45	172.33	664.52	-41.21	-17.91	7.87	1.048	0.337	-6.196
707.96	9.68	168.40	704.29	-47.81	-16.78	12.61	0.515	0.171	-2.923
748.28	11.47	163.50	743.92	-54.97	-14.96	18.25	1.489	1.332	-3.646
788.57	13.92	158.60	783.22	-63.33	-12.06	25.46	1.991	1.824	-3.649
828.91	15.68	154.79	822.22	-72.78	-7.96	34.27	1.495	1.309	-2.833
869.25	16.74	152.61	860.96	-82.87	-2.97	44.19	0.909	0.788	-1.621
909.60	16.06	153.68	899.67	-93.03	2.18	54.27	0.553	-0.506	0.796
949.94	14.95	155.21	938.54	-102.76	6.84	63.70	0.880	-0.825	1.138
990.29	13.70	155.13	977.63	-111.82	11.03	72.37	0.929	-0.929	-0.059
1,030.67	12.89	151.78	1,016.93	-120.13	15.17	80.56	0.830	-0.602	-2.489
1,071.00	12.15	152.10	1,056.30	-127.84	19.28	88.38	0.553	-0.550	0.238
1,111.37	12.64	147.23	1,095.73	-135.31	23.66	96.28	0.858	0.364	-3.619
1,151.73	15.84	143.99	1,134.85	-143.48	29.29	105.60	2.450	2.379	-2.408
1,192.07	18.18	149.72	1,173.42	-153.37	35.70	116.55	2.139	1.740	4.261
1,232.42	18.49	144.76	1,211.73	-164.03	42.57	128.33	1.182	0.230	-3.688
1,272.73	16.15	144.57	1,250.21	-173.82	49.51	139.66	1.742	-1.742	-0.141
1,313.08	15.42	145.64	1,289.04	-182.82	55.79	149.99	0.584	-0.543	0.796
1,353.43	15.54	145.81	1,327.92	-191.72	61.85	160.09	0.095	0.089	0.126
1,393.74	15.68	134.88	1,366.75	-200.04	68.75	170.53	2.188	0.104	-8.134
1,434.06	16.31	123.73	1,405.52	-207.03	77.32	181.56	2.330	0.469	-8.296
1,474.42	18.16	110.38	1,444.08	-212.36	87.93	193.31	3.237	1.375	-9.923
1,514.76	21.78	110.07	1,481.99	-217.12	100.86	206.60	2.693	2.692	-0.231
1,555.11	23.34	111.89	1,519.25	-222.67	115.31	221.59	1.271	1.160	1.353
1,595.46	24.60	113.51	1,556.12	-229.00	130.43	237.58	1.057	0.937	1.204
1,635.81	26.12	116.02	1,592.58	-236.25	146.11	254.57	1.384	1.130	1.866
1,676.14	25.75	111.05	1,628.86	-243.29	162.27	271.81	1.640	-0.275	-3.697
1,716.50	25.56	105.93	1,665.24	-248.83	178.83	288.52	1.653	-0.141	-3.806
1,756.82	25.59	106.13	1,701.61	-253.64	195.55	304.94	0.068	0.022	0.149
1,797.22	25.55	107.65	1,738.06	-258.70	212.24	321.47	0.488	-0.030	1.129
1,837.56	25.47	106.67	1,774.46	-263.83	228.84	337.97	0.319	-0.059	-0.729
1,877.90	25.62	106.64	1,810.86	-268.82	245.50	354.44	0.112	0.112	-0.022
1,918.25	25.57	108.44	1,847.25	-274.07	262.12	371.03	0.579	-0.037	1.338
1,958.61	25.50	107.99	1,883.67	-279.51	278.65	387.65	0.153	-0.052	-0.334
1,998.97	26.11	107.77	1,920.00	-284.90	295.37	404.40	0.459	0.453	-0.164
2,039.32	25.96	107.45	1,956.26	-290.26	312.25	421.26	0.153	-0.112	-0.238
2,079.67	26.64	109.21	1,992.43	-295.88	329.22	438.35	0.769	0.506	1.309

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Well:	F-4	North Reference:	Grid
Wellbore:	F-4	Survey Calculation Method:	Minimum Curvature
Design:	F-4	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,120.01	26.48	109.43	2,028.52	-301.85	346.24	455.68	0.140	-0.119	0.164	
2,160.35	26.41	110.00	2,064.63	-307.91	363.15	472.98	0.196	-0.052	0.424	
2,200.70	26.70	109.90	2,100.73	-314.07	380.11	490.37	0.218	0.216	-0.074	
2,241.06	26.47	109.31	2,136.82	-320.13	397.12	507.74	0.260	-0.171	-0.439	
2,281.44	26.44	106.54	2,172.97	-325.66	414.23	524.90	0.917	-0.022	-2.058	
2,321.84	26.41	106.65	2,209.15	-330.80	431.46	541.92	0.043	-0.022	0.082	
2,362.19	26.58	106.32	2,245.26	-335.90	448.72	558.94	0.167	0.126	-0.245	
2,402.55	26.67	106.08	2,281.34	-340.95	466.09	576.02	0.104	0.067	-0.178	
2,442.90	26.50	105.54	2,317.43	-345.87	483.47	593.04	0.220	-0.126	-0.401	
2,483.25	27.05	106.30	2,353.45	-350.86	500.95	610.17	0.482	0.409	0.565	
2,523.62	27.47	106.17	2,389.34	-356.03	518.70	627.64	0.315	0.312	-0.097	
2,563.96	27.40	105.47	2,425.14	-361.09	536.58	645.15	0.245	-0.052	-0.521	
2,604.31	26.84	107.02	2,461.06	-366.24	554.24	662.52	0.670	-0.416	1.152	
2,644.66	27.58	113.83	2,496.95	-372.68	571.50	680.32	2.378	0.550	5.063	
2,685.02	27.86	122.13	2,532.69	-381.47	588.04	698.89	2.875	0.208	6.169	
2,696.43	28.19	124.71	2,542.76	-384.43	592.51	704.25	3.303	0.868	6.784	
2,740.79	29.63	133.70	2,581.61	-397.97	609.06	725.59	3.093	0.974	6.080	
2,773.90	31.22	140.01	2,610.17	-410.21	620.50	742.00	3.232	1.441	5.717	
2,800.66	31.75	145.84	2,633.00	-421.35	628.91	755.31	3.464	0.594	6.536	
2,840.30	31.62	150.50	2,666.73	-439.03	639.88	774.50	1.855	-0.098	3.527	
2,879.85	32.11	155.23	2,700.33	-457.60	649.40	793.01	1.930	0.372	3.588	
2,920.07	32.76	160.49	2,734.28	-477.57	657.51	811.19	2.159	0.485	3.923	
2,959.08	34.44	162.51	2,766.77	-498.04	664.35	828.63	1.552	1.292	1.553	
3,001.10	35.31	165.32	2,801.25	-521.12	671.00	847.42	1.304	0.621	2.006	
3,041.46	36.60	171.51	2,833.93	-544.31	675.73	864.72	2.865	0.959	4.601	
3,081.29	37.88	178.79	2,865.65	-568.29	677.75	880.25	3.454	0.964	5.483	
3,121.75	40.87	183.49	2,896.93	-593.94	677.20	894.67	3.130	2.217	3.485	
3,162.81	43.81	186.08	2,927.28	-621.48	674.88	908.74	2.497	2.148	1.892	
3,203.53	46.88	187.49	2,955.90	-650.24	671.45	922.60	2.379	2.262	1.039	
3,244.08	50.42	190.19	2,982.68	-680.31	666.75	936.20	3.017	2.619	1.998	
3,284.75	52.56	193.10	3,008.01	-711.47	660.32	949.01	2.305	1.579	2.147	
3,325.09	54.85	195.14	3,031.89	-742.99	652.38	960.80	2.096	1.703	1.517	
3,365.11	55.23	196.13	3,054.82	-774.58	643.54	971.90	0.672	0.285	0.742	
3,404.82	55.15	197.06	3,077.49	-805.82	634.23	982.41	0.580	-0.060	0.703	
3,444.78	54.88	199.93	3,100.40	-836.86	623.85	991.94	1.777	-0.203	2.155	
3,484.98	54.67	202.72	3,123.59	-867.45	611.91	999.93	1.708	-0.157	2.082	
3,510.00	54.67	202.72	3,138.06	-886.28	604.02	1,004.42	0.000	0.000	0.000	

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Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m (As Installed)
Well:	F-4	North Reference:	Grid
Wellbore:	F-4	Survey Calculation Method:	Minimum Curvature
Design:	F-4	Database:	Production EDM P246N

Design Targets										
Target Name										
- hit/miss target	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude	
- Shape	(°)	(°)	(m)	(m)	(m)	(m)	(m)			
F4 Alternative 201106	0.00	0.00	2,984.90	-913.84	550.17	6,477,650.00	435,600.00	58° 26' 0.5694 N	1° 53' 49.7589 E	
- actual wellpath misses target center by 156.57m at 3458.60m MD (3108.36 TVD, -847.45 N, 619.91 E)										
- Point										
Volve F4 201006	0.00	0.00	2,999.90	-736.78	647.20	6,477,827.00	435,697.00	58° 26' 6.3425 N	1° 53' 55.5590 E	
- actual wellpath misses target center by 24.10m at 3302.31m MD (3018.57 TVD, -725.11 N, 657.03 E)										
- Point										
F4 50mSW of v11	0.00	0.00	2,999.90	-783.76	630.11	6,477,780.04	435,679.91	58° 26' 4.8153 N	1° 53' 54.5530 E	
- actual wellpath misses target center by 53.24m at 3343.85m MD (3042.67 TVD, -757.80 N, 648.31 E)										
- Point										
F4 100m SW of v11	0.00	0.00	2,999.90	-830.74	613.01	6,477,733.08	435,662.82	58° 26' 3.2881 N	1° 53' 53.5468 E	
- actual wellpath misses target center by 81.88m at 3385.11m MD (3066.23 TVD, -790.33 N, 638.91 E)										
- Point										
Volve I20	0.00	0.94	3,000.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E	
- actual wellpath misses target center by 40.47m at 3322.04m MD (3030.13 TVD, -740.58 N, 653.03 E)										
- Point										
Volve I21 Bunn	0.00	0.94	3,131.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E	
- actual wellpath misses target center by 73.91m at 3397.12m MD (3073.09 TVD, -799.77 N, 636.07 E)										
- Point										

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
251.20	251.20	30"	30.000	36.000	
1,351.90	1,326.45	13 3/8"	13.375	17.500	
2,760.50	2,598.67	9 5/8"	9.625	12.250	
3,508.00	3,136.91	7"	7.000	8.500	

Formations						
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)	
2,764.37	2,602.00	Tor	Empty	0.00		
2,940.05	2,751.00	Hod	Empty	0.00		
3,005.70	2,805.00	Blødøks	Empty	0.00		
3,041.55	2,834.00	Hidra	Empty	0.00		
3,055.29	2,845.00	Rødby	Empty	0.00		
3,066.58	2,854.00	Åsgard	Empty	0.00		
3,102.22	2,882.00	Draupne	Empty	0.00		
3,190.63	2,947.00	Heather	Empty	0.00		
3,250.88	2,987.00	Hugin	Empty	0.00		
3,430.15	3,092.00	Sleipner	Empty	0.00		
3,510.00	3,138.06	Total Depth	Empty	0.00		

Checked By: _____	Approved By: _____	Date: _____
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